

ACC_Ver02_Operating_Modes_ Requirements

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Requirement Set: ACC_Ver02_Operating_Modes_Requirements

Description

1 System Initialization

Requirement Type Functional

ID ACC-REQ-001

Description

On system start, the controller shall enter Power_Off with
cooling_demand = 0,
compressor_on = 0,
compressor_lockout = 0,
fan_on = 0,
fan_speed_cmd = 0,
and fault_indicator = 0.
Revision: Rev. 01
Change: Classification: Nil
Source Baseline: Ver. 01
Change Rationale: Nil

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Power Off](#) (←Implemented by)

2 Temperature Setpoint Input

Requirement Type Functional

ID ACC-REQ-002

Description

The controller shall use set_temperature as the desired room temperature reference for cooling control.
Revision: Rev. 01
Change: Classification: Nil
Source Baseline: Ver. 01
Change Rationale: Nil

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Power On](#) (←Implemented by)

3 Power ON Command

Requirement Type Functional

ID ACC-REQ-003

Description

When power_button == 1, the controller shall enter Power_On, set fault_indicator = 0, update set_temperature from desired_room_temperature, and compute temperature_error = room_temperature - set_temperature. The controller shall evaluate cooling demand only while cooling control is enabled.
Revision: Rev. 02
Change: Classification: Applicability clarification
Source Baseline: Ver. 01
Change Rationale: Power_On now also contains standby and Fan-only operation.

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Power On](#) (↔Implemented by)

 [\[power button == 1 \]](#) (↔Implemented by)

 [\[cooling control enabled == 1 \]](#) (↔Implemented by)

4 Power OFF Command

Requirement Type Functional

ID ACC-REQ-004

Description

When power_button == 0, the controller shall enter the Power_Off state and set compressor_on = 0 and fan_on = 0.

Revision: Rev. 01

Change: Nil

Source Baseline: Ver. 01

Change Rationale: Nil

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Power Off](#) (↔Implemented by)

 [\[power button == 0 \]](#) (↔Implemented by)

5 Cooling Demand Activation

Requirement Type Functional

ID ACC-REQ-005

Description

While cooling control is enabled, if temperature_error >= temperature_hysteresis_band, the controller shall set cooling_demand = 1.

Revision: Rev. 02

Change: Applicability clarification

Source Baseline: Ver. 01

Change Rationale: Cooling-demand hysteresis now applies only while cooling control is enabled.

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Cooling Demand Activation](#) (↔Implemented by)

 [\[temperature_error >= temperature_hysteresis_band \]](#) (↔Implemented

by)

 [\[cooling_control_enabled == 1 \]](#) (↔Implemented by)

6 Cooling Demand Deactivation

Requirement Type Functional

ID ACC-REQ-006

Description

While cooling control is enabled, if temperature_error <= -temperature_hysteresis_band, the controller shall set cooling_demand = 0.

Revision: Rev. 02

Change: Applicability clarification

Source Baseline: Ver. 01

Change Rationale: Cooling-demand hysteresis now applies only while cooling control is enabled.

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Cooling Demand Deactivation](#) (↔Implemented by)

 [\[temperature_error <= -temperature_hysteresis_band \]](#) (↔Implemented

by)

 [\[cooling_control_enabled == 1 \]](#) (↔Implemented by)

7 Compressor Minimum OFF Time

Requirement Type Functional

ID ACC-REQ-007

Description

After the compressor turns OFF, the controller shall prevent compressor restart until compressor_min_off_time has elapsed.

Revision: Rev. 01

Change: Nil

Source Baseline: Ver. 01

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Compressor Lockout](#) (↔Implemented by)

 [\[cooling_demand == 0 \]](#) (↔Implemented by)

 [\[after\(compressor_min_off_time, sec\) && cooling_demand == 1 \]](#)

(↔Implemented by)

 [Compressor Activation](#) (↔Implemented by)

 [\[after\(compressor_min_off_time, sec\) && cooling_demand == 0 \]](#)

(↔Implemented by)

 [Compressor Deactivation](#) (↔Implemented by)

8 Compressor Activation

Requirement Type Functional

ID ACC-REQ-008

Description

If cooling_demand == 1 and the compressor is not in minimum-OFF-time lockout, the controller shall set compressor_on = 1.

Revision: Rev. 01

Change: Nil

Source Baseline: Ver. 01

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Compressor Activation](#) (↔Implemented by)

 [\[after\(compressor min off time, sec\) && cooling_demand ==1 \]](#)

(↔Implemented by)

 [\[cooling_demand ==1 \]](#) (↔Implemented by)

9 Compressor Deactivation

Requirement Type Functional

ID ACC-REQ-009

Description

If cooling_demand == 0, power_button == 0, or a fault condition is active, the controller shall set compressor_on = 0.

Revision: Rev. 01

Change: Nil

Source Baseline: Ver. 01

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Compressor Deactivation](#) (↔Implemented by)

 [Power Off](#) (↔Implemented by)

 [Fault](#) (↔Implemented by)

 [\[power_button == 0 \]](#) (↔Implemented by)

 [\[fault == 1 \]](#) (↔Implemented by)

 [\[cooling_demand == 0 \]](#) (↔Implemented by)

10 Fan Activation During Compressor Activation

Requirement Type Functional

ID ACC-REQ-010

Description

When compressor_on == 1, the controller shall set fan_on = 1.

Revision: Rev. 01

Change: Nil

Source Baseline: Ver. 01

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Fan Activation](#) (↔Implemented by)

 [\[compressor_on == 1 \]](#) (↔Implemented by)

 [\[compressor_on == 1 \]](#) (↔Implemented by)

11 Fan Behavior After Compressor Deactivation

Requirement Type Functional

ID ACC-REQ-011

Description

When compressor_on == 0 after active cooling, the controller shall keep fan_on = 1 for configured fan_run_on_time, then set fan_on = 0.

Revision: Rev. 01

Change: Nil

Source Baseline: Ver. 01

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Compressor Off Check](#) (↔Implemented by)

 [Fan Deactivation](#) (↔Implemented by)

 [\[compressor_on == 0 \]](#) (↔Implemented by)

 [\[after\(fan_run_on_time, sec\) && compressor_on == 0 \]](#) (↔Implemented by)

by)

12 Default Power-On Standby Behavior

Requirement Type Functional

ID ACC-REQ-012

Description

When power_button == 1, no fault condition is active, and mode_enabled == 0, the controller shall activate Power_On_Standby, set cooling_demand = 0, compressor_on = 0, compressor_lockout = 0, fan_on = 1, and fan_speed_cmd = 1 corresponding to Low Speed. The controller shall set fan_only_mode_enabled = 0, cooling_mode_enabled = 0, and automatic_mode_enabled = 0.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Power On Standby](#) (↔Implemented by)

 [\[power_button == 1 \]](#) (↔Implemented by)

 [\[mode_enabled == 0 \]](#) (↔Implemented by)

13 Fan-Only Mode Selection and Response

Requirement Type Functional

ID ACC-REQ-013

Description

When power_button == 1, mode_enabled == 1, and mode_selected == 0, the controller shall activate Fan_Only_Mode, set fan_only_mode_enabled = 1, set cooling_mode_enabled

= 0 and automatic_mode_enabled = 0, disable cooling control, set cooling_demand = 0, compressor_on = 0, compressor_lockout = 0, and keep fan_on = 1. The controller shall determine fan_speed_cmd from fan_speed_user_input.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Fan Only Mode](#) (↔Implemented by)

 [\[power_button == 1 \]](#) (↔Implemented by)

 [\[mode_enabled == 1 && mode_selected == 0 \]](#) (↔Implemented by)

 [Manual Fan Speed Logic](#) (↔Implemented by)

14 Cooling Mode Selection and Response

Requirement Type Functional

ID ACC-REQ-014

Description

When power_button == 1, mode_enabled == 1, and mode_selected == 1, the controller shall activate Cooling_Mode, set cooling_mode_enabled = 1, set fan_only_mode_enabled = 0 and automatic_mode_enabled = 0, and enable the cooling-control mechanism. The controller shall determine cooling_demand using temperature_error and temperature_hysteresis_band, control the compressor using cooling demand and minimum-OFF-time protection, control the fan using compressor operation and fan run-on behavior, and determine fan_speed_cmd from fan_speed_user_input.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC Ver02 Operating Modes.slx](#)

 [Cooling Mode](#) (↔Implemented by)

 [\[power_button == 1 \]](#) (↔Implemented by)

 [\[mode_enabled == 1 && mode_selected == 1 \]](#) (↔Implemented by)

 [Cooling Control Mechanism](#) (↔Implemented by)

 [Manual Fan Speed Logic](#) (↔Implemented by)

15 Automatic Mode Selection and Response

Requirement Type Functional

ID ACC-REQ-015

Description

When power_button == 1, mode_enabled == 1, and mode_selected == 2, the controller shall activate Automatic_Mode, set automatic_mode_enabled = 1, set fan_only_mode_enabled = 0 and cooling_mode_enabled = 0, and enable the cooling-control mechanism. The controller

shall determine cooling demand automatically using the configured temperature logic. The controller shall select fan speed automatically when manual_fan_override == 0 and shall use manual fan-speed selection when manual_fan_override == 1.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

-  [Automatic Mode](#) (↔Implemented by)
-  [\[power button == 1 \]](#) (↔Implemented by)
-  [\[mode enabled == 1 && mode_selected == 2 \]](#) (↔Implemented by)
-  [Cooling Control Mechanism](#) (↔Implemented by)
-  [Manual Fan Speed Override State](#) (↔Implemented by)
-  [Automatic Fan Speed State](#) (↔Implemented by)
-  [\[manual fan override == 1 \]](#) (↔Implemented by)
-  [\[manual fan override == 0 \]](#) (↔Implemented by)
-  [Automatic Fan Speed Logic](#) (↔Implemented by)
-  [Manual Fan Speed Logic](#) (↔Implemented by)

16 Automatic Fan Speed Selection

Requirement Type Functional

ID ACC-REQ-016

Description

While Automatic_Mode is active and manual_fan_override == 0, the controller shall set automatic_fan_speed_enabled = 1 and determine fan_speed_cmd as follows:

If fan_on == 0, set fan_speed_cmd = 0.

If compressor_on == 1 and
temperature_error < medium_speed_threshold,
the controller shall set fan_speed_cmd = 1.

If compressor_on == 1 and
temperature_error >= medium_speed_threshold and
temperature_error < high_speed_threshold,
the controller shall set fan_speed_cmd = 2.

If compressor_on == 1 and
temperature_error >= high_speed_threshold,
the controller shall set fan_speed_cmd = 3.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

-  [Automatic Mode](#) (↔Implemented by)
-  [\[power button == 1 \]](#) (↔Implemented by)
-  [\[mode enabled == 1 && mode_selected == 2 \]](#) (↔Implemented by)
-  [\[manual fan override == 0 \]](#) (↔Implemented by)
-  [Automatic Fan Speed State](#) (↔Implemented by)
-  [Automatic Fan Speed Selection](#) (↔Implemented by)
-  [\[automatic fan speed enabled == 1 \]](#) (↔Implemented by)

17 Manual Fan Speed Selection and Override

Requirement Type Functional

ID ACC-REQ-017

Description

While Fan_Only_Mode or Cooling_Mode is active, or while Automatic_Mode is active with manual_fan_override == 1, the controller shall determine fan_speed_cmd from fan_speed_user_input.

If fan_on == 0, set fan_speed_cmd = 0.

If fan_speed_user_input == 0, set fan_speed_cmd = 1.

If fan_speed_user_input == 1, set fan_speed_cmd = 2.

If fan_speed_user_input == 2, set fan_speed_cmd = 3.

For any unsupported fan_speed_user_input, set fan_speed_cmd = 1 as the default fallback.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [\[\(fan only mode enabled == 1 || cooling mode enabled == 1 || automatic mode enabled == 1\) && automatic fan speed enabled == 0 \]](#)

(↔Implemented by)

-  [\[automatic fan speed enabled == 0 \]](#) (↔Implemented by)
-  [\[manual fan override == 1 \]](#) (↔Implemented by)
-  [Manual Fan Speed Override State](#) (↔Implemented by)
-  [Manual Fan Speed Selection](#) (↔Implemented by)

18 Compressor Lockout Status

Requirement Type Functional

ID ACC-REQ-018

Description

While the compressor minimum-OFF-time lockout is active, the controller shall set compressor_lockout = 1 and keep compressor_on = 0. After compressor_min_off_time has

elapsed, the controller shall set compressor_lockout = 0. The controller shall also set compressor_lockout = 0 when cooling control is disabled, during Power_Off, and during Fault.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Compressor Lockout](#) (↔Implemented by)

 [Compressor Activation](#) (↔Implemented by)

 [Cooling Control Standby](#) (↔Implemented by)

 [Power Off](#) (↔Implemented by)

 [Fault](#) (↔Implemented by)

 [\[after\(compressor_min_off_time, sec\) && cooling_demand == 1 \]](#)

(↔Implemented by)

 [\[cooling_control_enabled == 0 \]](#) (↔Implemented by)

 [\[power_button == 0 \]](#) (↔Implemented by)

 [\[fault == 1 \]](#) (↔Implemented by)

 [Compressor Deactivation](#) (↔Implemented by)

 [\[after\(compressor_min_off_time, sec\) && cooling_demand == 0 \]](#)

(↔Implemented by)

19 Fresh Cooling-Control Activation Lockout

Requirement Type Functional

ID ACC-REQ-019

Description

When cooling control transitions from disabled to enabled, the compressor mechanism shall start in minimum-OFF-time lockout with compressor_on = 0 and compressor_lockout = 1. Compressor activation shall not be permitted until compressor_min_off_time has elapsed and cooling demand is active.

Revision: Rev. 01

Change: New Ver. 02 requirement

Source Baseline: Ver. 02

Change Rationale: Nil

Links

Artifact [ACC_Ver02_Operating_Modes.slx](#)

 [Compressor Lockout](#) (↔Implemented by)

 [\[after\(compressor_min_off_time, sec\) && cooling_demand == 1 \]](#)

(↔Implemented by)

 [\[cooling_control_enabled == 1 \]](#) (↔Implemented by)

 [\[cooling_demand == 1 \]](#) (↔Implemented by)

 [Cooling Control Standby](#) (↔Implemented by)

Appendix

Artifact List

Simulink models:

#	Name	Folder	Version
1	ACC_Ver02_Operating_Modes.slx	D:\MBD-SILMIL_May_2026\MBD-Requirements_Based_Stateflow_Projects\AC Controller Project\ACC-Ver02-OperatingModes\model	1.4